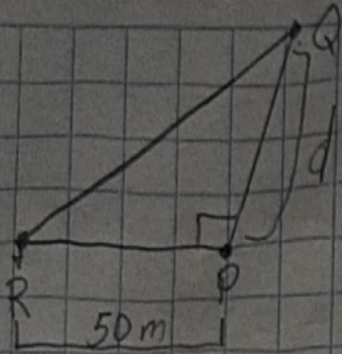


Tarea Sección 5.7

29)

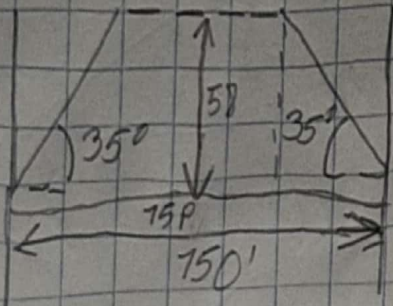


$$\tan 72^\circ 40' = \frac{50}{d}$$

$$d = 50(\tan 72^\circ 40') = 160.20$$

Res: La medida de d es de 160m.

33)



$$a) \sin 35^\circ = \frac{h}{75} \quad h = 75(\sin 35^\circ)$$

$$h = 43.02$$

$$h = 43.02 + 15 = 58.02$$

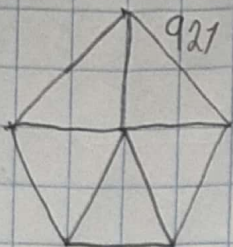
La distancia d es de 58 pies

$$b) \cos 35^\circ = \frac{x}{75}$$

$$x = 75(\cos 35^\circ) = 61.44 \times 2 = 122.88$$

$$150 - 122.88 = 27.12 \quad \text{Los extremos están a 27 pies}$$

41)



$$A = \frac{5L(A_p)}{2}$$

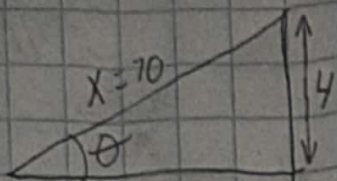
$$a = \frac{360^\circ}{5} = 72^\circ$$

$$A_p = \frac{L}{2 \tan \left(\frac{\theta}{2} \right)} = \frac{921}{2(0.7265)} = 633.83$$

$$A = \frac{5 \times 921 \times 633.83}{2} =$$

$$A = 1,459,379.5 \text{ m}^2 \text{ es la altura de la torre}$$

43)



$$X = \sqrt{(6)^2 + (8)^2}$$

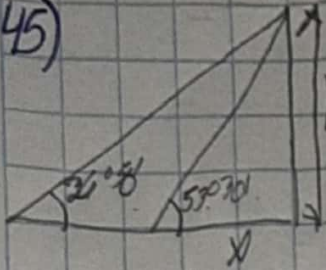
$$X = 10$$

$$\tan \theta = \frac{4}{10} \quad \theta = \tan^{-1}(4/10)$$

$$\theta = 21.8^\circ$$

El ángulo de la diagonal es de 21.8°

45)



$$\tan 26^\circ 50' = \frac{y}{25+x}$$

$$\tan 53^\circ 30' = \frac{y}{x}$$

$$x + \tan 53^\circ 30' = y$$

$$(25+x)(\tan 26^\circ 50') = y$$

$$(25+x)(\tan 26^\circ 50') = x \tan 53^\circ 30'$$

$$X = 14.26$$

$$25 \times \tan 26^\circ 50' = x \tan 53^\circ 30'$$

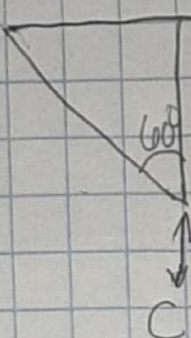
$$y = 14.26 (\tan 53^\circ 30')$$

$$26 \times \tan 36^\circ 30' = x$$

$$(\tan 53^\circ 30' - \tan 26^\circ 50') = x$$

$$y = 20.22 \text{ m es la altura de la torre}$$

55)

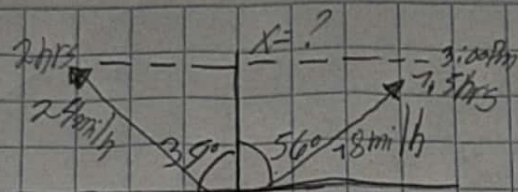


$$\text{sen } \alpha = \frac{h}{d}$$

$$h = d \text{ sen } \alpha$$

$$\underline{h = d \text{ sen } (\alpha) + c}$$

63) a)



$$d = vt$$

$$d = 24(2)$$

$$d = 48$$

$$d = 18(1.5)$$

$$d = 27$$

$$X = \sqrt{(48)^2 + (27)^2}$$

$$X = 55 \text{ miles a la distancia}$$

que encuentran a las 3:00 pm

$$b) \tan \theta = \frac{27}{48}$$

$$\theta = \tan^{-1} \frac{27}{48}$$

$$\theta =$$

73) a) $h = 50$

$$p = 30$$

$$v = 180 \text{ ft/s}$$

$$a = 25$$

$$b = \frac{2\pi}{p} = \frac{2\pi}{30} = \frac{\pi}{15}$$

$$y = 25 \cos \frac{\pi}{15} t$$

b) $d = vt$

$$\frac{180 \text{ ft}}{1 \text{ s}} \times \frac{30 \text{ mi}}{1 \text{ min}} \times \frac{60 \text{ seg}}{1 \text{ min}} = 324,000 \text{ ft es la longitud de onda}$$